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Preamble

Everything looks normal above the desk and that's what counts.

Kevin Hart

Welcome to *How to Pass Inorganic Chemistry*. The content of this book is designed to help you prepare for the UACE Chemistry Examination. It provides a complete summary of the Advanced Inorganic Chemistry and offers the following features to aid understanding and improve recall of important points.

Example: In chapter 1 and 6 of this book, you will find sets of solved examples to show you how to answer the questions. It is essential for your learning to make every effort to recall information; this effort of mentally searching for an answer helps to strengthen your memory and will boost your ability to learn. Review your answer against the exemplar provided.

Knowledge Check: These are short questions at the end of a section/concept/sample problem that test your understanding of the concept to allow you apply the knowledge and skills you have acquired.

Study Tip: Study tips contain prompts to help you with your understanding and revision. This feature gives you extra hints and tips and draws your attention to particular things to think about and remember.

Exam-style Question: Exam-style questions are found at the end of each chapter except in chapter 6. They are examination-like questions and should be used to test your knowledge and understanding of the chemistry covered. The questions included have been selected carefully and some are modelled on those found in previous UACE examinations. Like the knowledge check, you should always try to answer the question using your own recall of the topic.

How to pass inorganic chemistry pays attention to what the examiners want as seen in the study tips, knowledge check, examples and Exam-style questions.

I hope that *How to Pass Inorganic Chemistry* will help you to make good progress in inorganic chemistry and that you will enjoy it.

Suggestions and comments of readers, both students and facilitators will be sincerely welcome and appreciated.

Charles Kidega

Contents

Preamble	ii
1 Periodicity	1
1.1 Electrons in atoms	1
1.2 Atomic radius	3
1.3 Ionic radius	5
1.4 Ionisation energy	7
1.5 Electron affinity	10
1.6 Electronegativity	13
1.7 Electropositivity	14
1.8 Melting and boiling points	16
2 Alkaline Earth Metals	21
2.1 Introduction to group II elements	21
2.2 Reactions of group II elements	23
2.3 Compounds of group II elements	26
2.4 Anomalous behavior of beryllium	28
2.5 Chemical resemblance of magnesium	29
2.6 Qualitative analysis of group II cations	30
3 Group IV Elements	36
3.1 Introduction to group IV elements	36
3.2 Reactions of group IV elements	38
3.3 Compounds of group IV elements	41
3.4 Anomalous behavior of carbon	47
3.5 Qualitative analysis of lead(II) ions	48
4 Group VII Elements	56
4.1 Introduction to group IV elements	56
4.2 Reactions of group IV elements	59
4.3 Compounds of group IV elements	61
4.4 Anomalous behavior of fluorine	64
4.5 Qualitative analysis of halide ions	65
5 Period 3 Elements	71
5.1 Introduction to period 3 elements	71
5.2 Reactions of period 3 elements	72

5.3	Compounds of period 3 elements	76
5.4	The chemistry of aluminium	81
6	Transition Elements	89
6.1	Introduction to transition metals	89
6.2	Characteristics of the transition metals	91
6.3	Chemistry of chromium	97
6.4	Chemistry of manganese	104
6.5	Chemistry of iron	111
6.6	Chemistry of cobalt	118
6.7	Chemistry of nickel	121
6.8	Chemistry of copper	123
6.9	Chemistry of zinc	131
7	Industrial chemistry	135
7.1	Extraction of metals	135
7.2	Manufacture of sulphuric acid	138
7.3	Manufacture of nitric acid	139
7.4	Manufacture of sodium hydroxide	140
8	Qualitative analysis	142
8.1	Introduction to qualitative analysis	142
8.2	Preliminary tests	143
8.3	Tests for cations	145
8.4	Tests for anions	148
	Quick Navigation	151

- (iii) To which group in the Periodic Table does element B belong? Give a reason for your answer.
- (b) (i) Explain what is meant by the term *electronegativity*.
 (ii) State the factors that determine the value of electronegativity of an element.
3. The table below shows the first three ionisation energies in kJ mol^{-1} of the elements A, B, C and D.

Element	Ionisation energies (kJ mol^{-1})		
	First	Second	Third
A	730	1500	7730
B	500	4560	6900
C	580	1815	2740
D	1310	3460	5300

- (a) Which one of the elements belongs to group I of the Periodic Table? Give a reason for your answer.
- (b) Which one of the elements forms a cation with three positive charges?
- (c) Write the formula of the compound formed by the elements A and D.
4. The first successive ionization energies of elements X and Y are given in the table below.

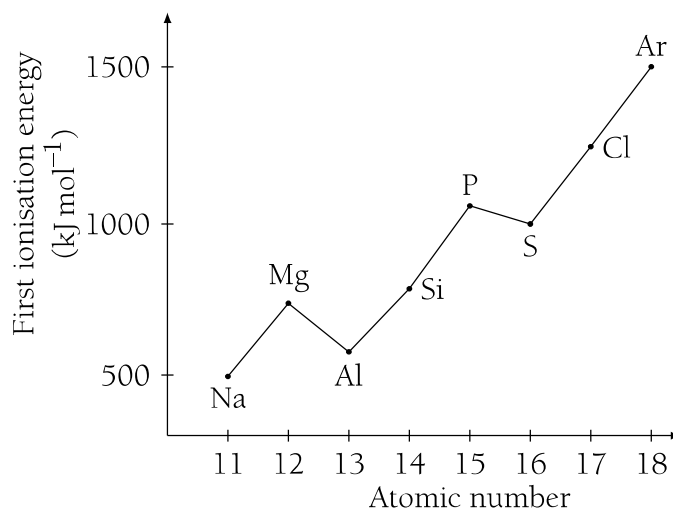
Element	Ionisation energies (kJ mol^{-1})				
	First	Second	Third	Fourth	Fifth
X	1310	3390	5320	7460	11000
Y	1410	2860	4710	7470	9450

- (a) State the group in the Periodic Table which each of the elements belongs to. Explain your answer.
- (b) Elements A and B both have high ionization energies and high electron affinities. State the type of bond that exists in the compound formed by A and B.
5. (a) (i) What is meant by the term *ionization energy* of an element?
 (ii) Write an equation to show the first ionization energy of magnesium.
- (b) The second and third ionization energies of magnesium are 1450 and 7730 kJ mol^{-1} respectively. Give a reason for the large difference between the second and third ionization energies of magnesium.

6. The first four ionization energies (kJ mol^{-1}) of some elements are shown in the table below.

Element	1st	2nd	3rd	4th
A	500	4600	9600	9500
B	740	1500	7700	10500
C	630	1600	3000	4800
D	900	1800	14800	21000
E	580	1800	2700	11600

- (a) What is meant by the term *first ionization energy*?
- (b) State the element that is most likely to form an ion with unit positive charge. Give a reason for your answer.
- (c) Identify
- (i) two elements that are in the same group of the Periodic Table.
 - (ii) the group to which the elements you have stated in (c)(i) belong.
- (d) Write
- (i) the formula of the chloride of element E.
 - (ii) equation for the reaction between the chloride of E and water.
7. (a) The graph of first ionisation energy against atomic number for the elements in the third short period of the Periodic Table is shown below.

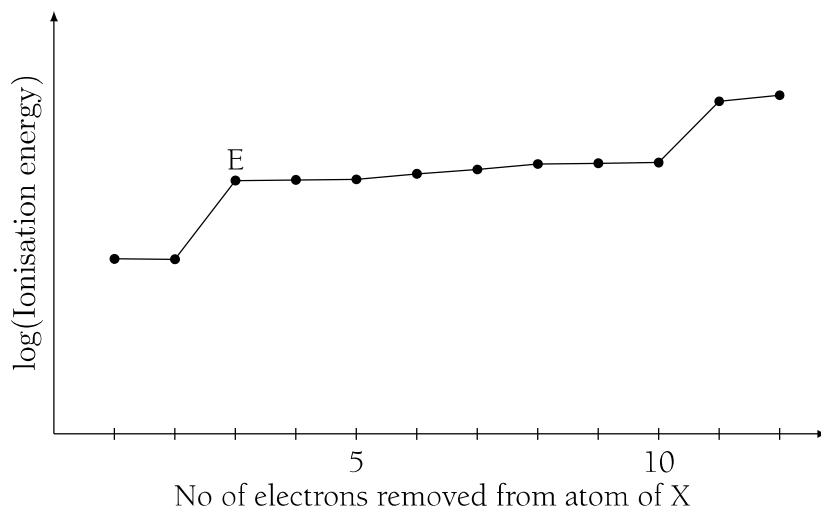


- (i) What is meant by the term *ionisation energy*? Illustrate your answer using sodium atom.
- (ii) Explain the general trend in the first ionisation energy of the elements shown in the group.
- (b) Explain why the first ionisation energy of
- (i) aluminium is less than that of magnesium.
 - (ii) sulphur is less than that of phosphorus.

8. The first three successive ionisation energies of element W are 549, 1064 and 4226 kJ mol⁻¹.
- Explain the trend in the variation of the ionisation energies of W.
 - State the group in the Periodic Table to which W belongs.
9. The table below shows the atomic radius and the first ionisation energy of some elements in period 3 of the Periodic Table.

Element	Na	Mg	Al	Si	P	S	Cl
Atomic radius (nm)	0.186	0.160	0.143	0.117	0.110	0.104	0.099
First ionisation energy (kJ mol ⁻¹)	502	745	587	791	1020	1000	1260

- State how atomic radius of the elements varies across the period.
 - Explain your answer in (a)(i).
 - Explain how atomic radius affects the first ionisation energy.
 - Why is the first ionisation energy of aluminium lower than that of magnesium?
10. The diagram below shows successive ionization energies for an element X, showing removal of all electrons.



- Giving reasons, state
 - the group of element X
 - the period of element X
 - Identify element X
- Explain the sudden increase in the energy required to remove electron E.
- Explain how the size of X will change as electrons are removed.
- Explain what would be the sign of energy change if an electron was added to X to give X⁻.
 - How would you expect it to affect the size of X.